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B Discussions and Broader Impact

In Sec. B.1 we discuss the broader impact of our work and in Sec. B.2 limitations and future work.

B.1 Broader Impact

`BigEarthNet.txt` provides a foundation to advance the development of VLMs for the complex spatial and spectral content of RS. By providing co-registered multi-sensor data with diverse text annotations describing complex LULC patterns `BigEarthNet.txt` can serve as a valuable pre-training resource. The manually-checked `BigEarthNet.txt` benchmark split on the other hand enables the faithful evaluation of pre-trained VLMs. Further, through `BigEarthNet.txt` natural language interaction with RS archives can be made possible thus reducing the barrier to entry for non-experts. Eventually, this would allow the broader public to assess and analyze RS archives and therefore accelerate and democratize the countless applications. Finally, as we will publicly release the complete dataset generation pipeline the overall process could easily be adapted and applied to different sources even beyond RS. The generation of almost 500 000 LLM-augmented and self-refined captions as well as the fine-tuning of RS-InternVL and model inference for SOTA analysis incurs non-trivial computational cost and energy consumption. As the dataset will be publicly released this is a one-time cost.

B.2 Limitations and Future Work

While `BigEarthNet.txt` will prove a valuable resource for advancements in image-text learning for RS, it has some limitations. `BigEarthNet.txt` is limited in geographical, coverage to ten European countries in terms of CLC LULC classes. Consequently, LULC categories that are foreign to Europe or not included in CLC are not covered. At the moment the co-registered multi-sensor data is limited to two sensors (S1 and S2). Further, we employ LLMs for rephrasing and CLAIR evaluation which could incur hallucinations or unfactual responses. Our benchmark split counters this as it is manually-verified and completely correct. For captioning, we also provide multiple other standard captioning metrics. Finally, while the text annotations of `BigEarthNet.txt` are already

diverse in semantic content as well as type there exists an inherent limit due to the generation from CLC maps and the 19-class nomenclature. As future work we plan to increase geographical coverage of `BigEarthNet.txt`, consider additional LULC sources, add additional sensing modalities, and increase annotation type variety even further.

C Prompts for Caption Generation Pipeline

Figure 1 shows the full prompt used for LLM-based paraphrasing of template-based captions. The prompt comprises a general task-description, set of rules to adhere to, in-context examples (shortened for readability), and the actual template-based caption input. Finally we provide the rephrased output from the model.

Prompt for the LLM-based Augmentation - Paraphrase

Task Description

You are a remote-sensing expert. You will be given a static description of a satellite image with well-defined classes. The well-defined classes are enclosed in quotation marks, followed by their area in square meters in brackets. The description also includes information about which classes are bordering or adjacent to one another, ordered by their relative size without repetition. Use varying natural language and sentence structure to rephrase the description into a single, cohesive, well-written remote sensing caption that contains all of the provided information.

Use the following hierarchy definition to further increase the variance in your output:

- "Urban fabric" and "Industrial or commercial units" are part of the more general class "Artificial surfaces".
- "Arable land", "Permanent crops", "Pastures", "Complex cultivation patterns", "Land principally occupied by agriculture with significant areas of natural vegetation", and "Agro-forestry areas" are part of the more general class "Agricultural areas".
- "Broad-leaved forest", "Coniferous forest", "Mixed forest", "Natural grasslands or sparsely vegetated areas", "Moors, heathland or sclerophyllous vegetation", "Transitional woodlands or shrubs", and "Beaches, dunes, or sands" are part of the more general class "Forest and semi-natural areas".
- "Coastal wetlands" and "Inland wetlands" are part of the more general class "Wetlands".
- "Inland waters" and "Marine waters" are part of the more general class "Water bodies".

Set of Rules

While writing the caption, do not violate the following rules:

- Use precise language to describe observable features.
- Avoid assumptions or interpretations beyond the given data.
- Do not make assumptions towards the land use and avoid the phrases "land-cover" and "land-use".
- Include the associated area information if you do not change the class names.
- If you group classes, ensure to either remove the associated areas or sum them together and re-evaluate the border information.
- Either choose 5 % or 72'000 sqm when talking about the smaller regions, but do not use both.
- Remove the quotation marks for the predefined classes in your output.
- Do not prefix your output with "Output:".
- If the input mentions the associated climate zone, keep the climate zone definition as is.
- Include the seasonal and country information in the output.

In-context Examples	The following provides a few examples. Input: The image primarily shows “Arable land” (∼708’000 sqm), followed by “Broad-leaved forest” (∼413’000 sqm). Smaller areas are covered by “Pastures” (∼319’000 sqm) and “Mixed forest” (∼204’000 sqm). “Arable land” borders “Broad-leaved forest” and “Pastures”. Moreover, “Broad-leaved forest” is adjacent to “Mixed forest”. The “Arable land” is distributed over three individual areas, where one is a larger area (∼525’000 sqm) and two are smaller areas (110’000 sqm and 73’000 sqm). The “Mixed forest” is distributed over two individual smaller areas (116’000 sqm and 88’000 sqm). The image was taken during Summer in Austria. The associated climate zone is “cold, no dry season, warm summer”. Output: This satellite image, captured in Austria during summer, depicts a diverse landscape dominated by agricultural and forested areas. Arable land (708,000 sqm) is most prominent, covering a large area and two smaller areas, and is next to broad-leaved forest (∼413,000 sqm) and pastures (∼319,000 sqm). The expansive broad-leaved forest is adjacent to a mixed forest (∼204,000 sqm), which is spread across two smaller areas, suggesting a varied forest ecosystem. The landscape is characterized by a mix of agricultural, forest and seminatural areas, set within the “cold, no dry season, warm summer” climate zone. Input: ... Output: ... Input: ... Output ...
Input Template Caption	The image primarily shows “Arable land” (∼526’000 sqm), followed by “Inland wetlands” (∼460’000 sqm). Smaller areas are covered by “Inland waters” (∼305’000 sqm) and “Urban fabric” (∼149’000 sqm). “Arable land” borders “Inland wetlands” and “Urban fabric”. Moreover, “Inland wetlands” is adjacent to “Inland waters” and “Urban fabric”. The “Urban fabric” is distributed over three individual marginal areas. The image was taken during Summer in Switzerland. The associated climate zone is “temperate, no dry season, warm summer”.

This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the “temperate, no dry season, warm summer” climate zone. The dominant features are arable land (∼526,000 sqm) and inland wetlands (∼460,000 sqm), which are adjacent to each other. The arable land also borders urban fabric (∼149,000 sqm), which is scattered across three small, marginal areas. Furthermore, the inland wetlands are next to inland waters (∼305,000 sqm) and urban fabric. The varied landscape presents a mix of agricultural areas, wetlands, water bodies, and artificial surfaces.

Fig. 1: Promp for LLM-based caption rephrasing based on template-based version, including general task description, set of rules for the model to follow, three in-context examples (shortened for brevity), actual template-based caption input, and model output. As model we employ LLama-4-Scout-17B [36].

In Fig. 2 we show the prompt for LLM-based caption refinement. This includes the paraphrase prompt and output from Fig. 1, followed by a refinement instruction asking the model to improve the paraphrased caption based on the initial template-based version and rephrasing instruction.

LLM-based Augmentation - Refinement

Paraphrase Prompt You are a remote-sensing expert. You will be given a static description of a satellite image with well-defined classes. The well-defined classes are enclosed in quotation marks, followed by their area in square meters in brackets. The description also includes information about which classes are bordering or adjacent to one another, ordered by their relative size without repetition. Use varying natural language and sentence structure to rephrase the description into a single, cohesive, well-written remote sensing caption that contains all of the provided information.
...

This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant features are arable land (∼526,000 sqm) and inland wetlands (∼460,000 sqm), which are adjacent to each other. The arable land also borders urban fabric (∼149,000 sqm), which is scattered across three small, marginal areas. Furthermore, the inland wetlands are next to inland waters (∼305,000 sqm) and urban fabric. The varied landscape presents a mix of agricultural areas, wetlands, water bodies, and artificial surfaces.

Refinement Instruction Compare your rephrased caption with the initially provided static description. Ensure that your output fulfills the previously defined rules and does not have less information than the original description. Update the caption accordingly and provide the full caption in your output. Do NOT explain what mistake you have found or what you are correcting. Only include the updated caption in your response.

This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant features are arable land (∼526,000 sqm) and inland wetlands (∼460,000 sqm), which are adjacent to each other. The arable land borders both inland wetlands and urban fabric (∼149,000 sqm). Moreover, the inland wetlands are adjacent to both inland waters (∼305,000 sqm) and urban fabric. Notably, the urban fabric is distributed over three individual marginal areas. The varied landscape presents a mix of agricultural areas, wetlands, water bodies, and artificial surfaces.

Fig. 2: Prompt used for LLM-based caption self-refinement, including paraphrasing prompt with output from Fig. 1, the refinement instruction and the refined caption output from the model.

D Instruction Templates

We present examples of the instruction templates used for each task from; i) binary VQA in Tab. 1, ii) multiple-choice VQA in Tab. 2, iii) referring expression detection in Tab. 3, and iv) captioning in Tab. 4. Overall 469 templates are used for `BigEarthNet.txt`.

Table 1: Examples of question templates for each task of the binary VQA category. {...} denotes a placeholder for the actual value depending on the image content.

Task	**Templates**
Presence	Is/Are any {class} present in the satellite image? Is/Are any {class} shown in the image? Do you see any area(s) of {class} in the satellite image? Are there regions of {class} in the satellite image? Is any portion of the image covered by {class}?
Area	Is the image fully covered by {class}? Do {class} cover {bound} of the image? Are {class} occupying {lower bound} or more of the image? Is the total area of {class} less than {upper bound}? Do {class} cover between {lower bound} and {lower bound} of the image?
Count	Does the image contain exactly one continuous region of {class}? Are {class} present in multiple distinct continuous areas within the image? Does the image show exactly two distinct continuous areas of {class}? Do {class} occupy two or less continuous areas in the image? Would you say that {class} occupy four or more continuous areas in the image?
Adjacency	Does any (instance of) {class} border a/_ {class} in this scene? Is there any (instance of) {class} and a/_ {class} directly connected in the scene? Is any (instance of) {class} directly adjacent to a/_ {class} in this image? Is there direct adjacency between any (instance of) {class} and a/_ {class} in the scene? Does any (instance of) {class} directly border a/_ {class} in this scene?

Table 2: Examples of question templates for each task of the MCQ category. {...} denotes a placeholder for the actual value depending on the image content.

Task	Templates
Presence	Which of these classes is present in the image? a) {class}, b) {class}, c) {class}, d) {class} Which land cover class appears in this satellite image? a) {class}, b) {class}, c) {class}, d) {class} From these options, which class can be observed in the image? a) {class}, b) {class}, c) {class}, d) {class}
Area	How much of the image do {class} cover? a) {area range}, b) {area range}, c) {area range}, d) {area range} What is the total area of {class}? a) {area range}, b) {area range}, c) {area range}, d) {area range} Which range corresponds to the area covered by {class}? a) {area range}, b) {area range}, c) {area range}, d) {area range} How large is the area covered by {class}? a) {area range}, b) {area range}, c) {area range}, d) {area range}
Count	How many continuous areas are covered by {class}? a) {number}, b) {number}, c) {number}, d) {number} Count the separate regions of {class}: a) {number}, b) {number}, c) {number}, d) {number} How many separate areas of {class} can be seen? a) {number}, b) {number}, c) {number}, d) {number}
Adjacency	Identify the adjacent classes: a) {class pair}, b) {class pair}, c) {class pair}, d) {class pair} Which classes are directly connected? a) {class pair}, b) {class pair}, c) {class pair}, d) {class pair} Which pair of classes share a boundary? a) {class pair}, b) {class pair}, c) {class pair}, d) {class pair}
Relative Position	How is/are the {class} relatively positioned to the {class} in the image? a) {rel. pos.}, b) {rel. pos.}, c) {rel. pos.}, d) {rel. pos.} In relation to the {class}, where is/are the {class} located in the image? a) {rel. pos.}, b) {rel. pos.}, c) {rel. pos.}, d) {rel. pos.} What is the relative position of the {class} to the {class}? a) {rel. pos.}, b) {rel. pos.}, c) {rel. pos.}, d) {rel. pos.}
Country	Which country is shown in the satellite image? a) {country}, b) {country}, c) {country}, d) {country} From the options below, select the country shown in the image: a) {country}, b) {country}, c) {country}, d) {country} Choose the country depicted in the image: a) {country}, b) {country}, c) {country}, d) {country}
Season	In which of the following seasons was the satellite image taken? a) {season}, b) {season}, c) {season}, d) {season} Select the season shown in this satellite image: a) {season}, b) {season}, c) {season}, d) {season} In which season was the satellite image captured? a) {season}, b) {season}, c) {season}, d) {season}
Climate Zone	From the following options, pick the one that corresponds to the climate zone shown in the satellite image: a) {climate zone}, b) {climate zone}, c) {climate zone}, d) {climate zone} Identify which of the following climate zones is shown in the satellite image: a) {climate zone}, b) {climate zone}, c) {climate zone}, d) {climate zone} Which climate zone does the satellite image capture? a) {climate zone}, b) {climate zone}, c) {climate zone}, d) {climate zone}

Table 3: Examples of question templates for referring LULC detection and referring point detection. {...} denotes a placeholder for the actual value depending on the image content.

Task	Templates
Referring LULC Detection	Where can the smallest/largest continuous region of {class} be found? Identify the location of the smallest/largest contiguous area of {class}. Where is the smallest/largest continuous area of {class} in the image? Locate a continuous area of {class}. Indicate the position of a continuous region of {class}.
Referring Point Detection	Output a bounding box surrounding the land cover class instance positioned at {(x, y)} in the image. Generate a bounding box enclosing the land cover class instance at {(x, y)} in the satellite image. Create a bounding box around the land cover class instance found at {(x, y)} in the satellite image. Provide a bounding box enclosing the land cover class instance located at {(x, y)} in the image. Output a bounding box surrounding the land cover class instance at {(x, y)} in the satellite image.

Table 4: Examples of question templates for captioning.

Task	Templates
Captioning	Describe this satellite image, noting the region, seasonal context, and distribution of land cover types. Provide a comprehensive description of the scene, including the location, seasonal context and land cover distribution. Describe the observed landscape, specifying the geographic location, season, and land cover composition. Explain what this image shows, including the region, climate zone, and land cover characteristics. Provide an description of the image content, highlighting the geographic location, season, and spatial patterns.

E Additional Results on Test Set

Additional results for the main categories on the full `BigEarthNet.txt` test set are detailed in Tab. 5. Due to computational constraints we evaluate the strongest models from RS (EarthMind) and CV (Qwen3-VL), the backbone model for RS-InternVL (InternVL-3 (1B)), and the fine-tuned RS-InternVL.

Table 5: Inference performance of RS and CV VLMs on the main tasks on the `BigEarthNet.txt` test split. Reported metrics for captioning: BLEU-4, binary VQA: accuracy, MCQ: accuracy, and referring expression detection: mIoU. All results are in percent (%).

Model	# Parameters	Captioning	Binary VQA	MCQ	Ref. Exp. Det.
$\text{EarthMind}_{\text{s1s2}}$ [27]	4B	1.59	55.36	34.73	16.25
Qwen3-VL [1]	8B	0.78	61.27	37.06	18.60
InternVL-3 (1B) [45]	1B	0.52	51.44	27.56	5.94
RS-InternVL (ours)	1B	**31.06**	**80.06**	**74.42**	**51.35**

From the table one can observe that performance on the full test set is mostly similar (within a few %) compared to performance on the `BigEarthNet.txt` benchmark split. The deviation in performance is mostly due to the sampling strategy employed for the benchmark split to allow unbiased and fair evaluation. Accordingly, RS-InternVL, fine-tuned on the train set shows the largest performance difference. For fair, unbiased, manually-checked evaluation we recommend evaluating on the benchmark split which also drastically reduces computational cost.

F Full Instructions for Model Evaluation on `BigEarthNet.txt`

Table 6 lists the precise prompts per category used for model evaluation. "{Instruction}" denotes the instruction text directly from `BigEarthNet.txt`. We generally only add additional prompts when required. This includes model-specific special tokens (*e.g.* `[refer]` for GecoChat), or instructions to follow the expected output format (*e.g.* `"Answer with a single word:"yes" or "no"."` when the model alone is unable to reliably do so. Additionally, for area VQAs as well as for the captioning task we add `"The whole image area is 1440000 square meters."` before the instruction, to enable the relation between pixels and areas for each model.

Table 6: Per model per category prompt used during inference. {Instruction} denotes the instruction text directly from `BigEarthNet.txt`.

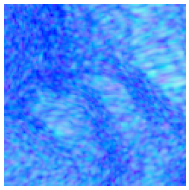
Model	Binary VQA	MCQ	Ref. Exp. Det.	Captioning
GeoChat	{Instruction} Answer with a single word: "yes" or "no".	{Instruction} Answer with exactly one of a, b, c, d.	[refer] {Instruction}	{Instruction}
LHRS-Bot	{Instruction} Answer with a single word: "yes" or "no".	{Instruction} Answer with exactly one of a, b, c, d.	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Coordinates must be in the 224x224 image space. Do not include any other text.	{Instruction}
SkyEyeGPT	[vqa] {Instruction} Answer with a single word: "yes" or "no".	[vqa] {Instruction} Answer with exactly one of a, b, c, d.	[refer] {Instruction}	[caption] {Instruction}
ED_{RGB}	[s2_rgb_10] {Instruction} Answer with a single word: "yes" or "no".	[s2_rgb_10] {Instruction}	[refer] [s2_rgb_10] {Instruction}	[caption] [s2_rgb_10] {Instruction}
ED_{MS}	[s2_ms_10] {Instruction} Answer the question using a single word.	{Instruction} Answer with exactly one of A, B, C, or D.	[refer] [s2_ms_10] {Instruction}	[caption] [s2_ms_10] {Instruction}
EarthMind	{Instruction} Answer with a single word:"yes"or "no".	{Instruction} Answer with exactly one of a, b, c, d.	{Instruction}	{Instruction}
gpt-5.2	{Instruction} Answer with a single word: "yes" or "no".	{Instruction} Answer using either a), b), c), or d).	{Instruction} Answer by only providing the bounding box in the format of: [x_min, y_min, x_max, y_max] where (x_min, y_min) is the top-left corner and (x_max, y_max) is the bottom-right corner and all values are between 0 and 1. The origin of the image is at the top-left corner.	{Instruction} Answer in form of a descriptive, unformatted caption."
Qwen	{Instruction}	{Instruction}	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Values in the ranges of 0 to 1.	{Instruction}
GLM	{Instruction}	{Instruction}	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Values in the ranges of 0 to 1.	{Instruction}
LLaVA	{Instruction}	{Instruction}	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Values in the ranges of 0 to 1.	{Instruction}
RS-InternVL	{Instruction}	{Instruction}	{Instruction}	{Instruction}

G Qualitative Examples of Model Answers

Figures 3 to 5 show qualitative per model answers for the binary VQA, MCQ, and referring expression detection tasks. Figure 6 gives an example for the captioning task with RS model answers and Fig. 7 shows the answers of CV models. One can clearly see that given adequate pre-training data a model is capable of answering correctly. Especially in Figs. 6 and 7 it is evident that multi-sensor and diverse pre-training data leads to a more specific description of complex LULC classes (*e.g.* "mixed forest" instead of just "forest"). The considered RS models are GeoChat [15] (denoted as GC), LHRS-Bot [24] (denoted as LHRS), SkyEyeGPT [42] (denoted as SE), EarthDial [30] (denoted as ED_{rgb}/ ED_{s2}), and EarthMind [27] (denoted as EM_{rgb}/ EM_{s1s2}). The considered CV models are GPT-5.2 [29] (denoted as GPT), Qwen3-VL [1] (denoted as Qwen), GLM-4.6v-Flash [35] (denoted as GLM), LLaVa-OneVision [16] (denoted as LLaVa), and InternVL3-1B [45] (denoted as IVL). Further we provide model responses for the fine-tuned RS-InternVL (denoted as IVL_{RS}).

Do regions of any coniferous forest directly adjoin a mixed forest?

S2

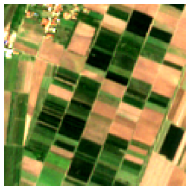


S1

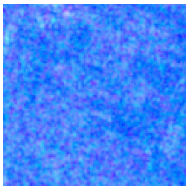
Model	Answer	Correct
GC	yes	✗
LHRS	yes	✗
SE	yes	✗
ED_{rgb}	no	✓
ED_{s2}	yes	✗
EM_{rgb}	yes	✗
EM_{s1s2}	yes	✗
GPT	yes	✗
Qwen	yes	✗
GLM	yes	✗
LLaVa	yes	✗
IVL	no	✓
IVL_{RS}	no	✓

Reference map

Is there only a single continuous region of urban fabrics visible in the image?

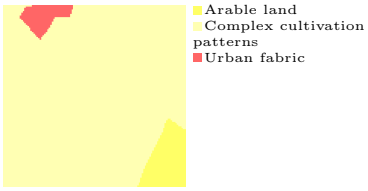


S2



S1

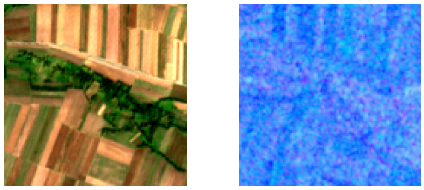
Model	Answer	Correct
GC	no	✗
LHRS	yes	✓
SE	yes	✓
ED_{rgb}	yes	✓
ED_{s2}	no	✗
EM_{rgb}	no	✗
EM_{s1s2}	no	✗
GPT	no	✗
Qwen	no	✗
GLM	yes	✓
LLaVa	no	✗
IVL	no	✗
IVL_{RS}	yes	✓



Reference map

Fig. 3: Qualitative examples for binary VQA capabilites of considered models. The S2 input is visualized as the true color RGB representation of the S2 image and the S1 input is visualized as (RGB: VV, VH, VV/VH). We provide the reference maps as aid to the reader only. Reference maps are not provided as input to the model. "–" indicates that the answer could not be extracted.

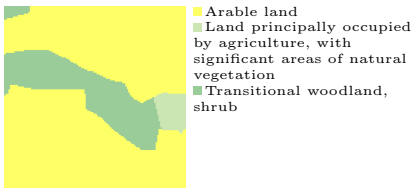
Identify the country depicted in the satellite image:



S2 S1

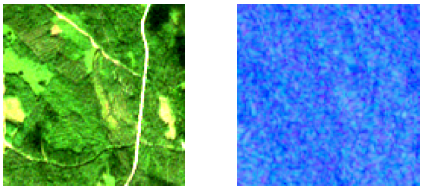
a) Austria
b) Switzerland
c) Luxembourg
d) Serbia

Model	Answer	Correct
GC	a	✗
LHRS	d	✓
SE	c	✗
ED_{rgb}	a	✗
ED_{s2}	a	✗
EM_{rgb}	c	✗
EM_{s1s2}	d	✓
GPT	c	✗
Qwen	a	✗
GLM	c	✗
LLaVa	b	✗
IVL	b	✗
IVL_{RS}	d	✓



Reference map

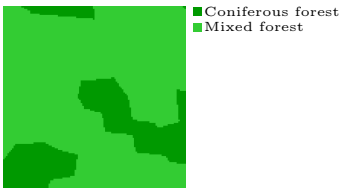
From the following classes, select the one that is present:



S2 S1

a) Broad-leaved forest
b) Coniferous forest
c) Coastal wetlands
d) Inland wetlands

Model	Answer	Correct
GC	a	✗
LHRS	a	✗
SE	a	✗
ED_{rgb}	–	✗
ED_{s2}	d	✗
EM_{rgb}	–	✗
EM_{s1s2}	a	✗
GPT	a	✗
Qwen	a	✗
GLM	a	✗
LLaVa	a	✗
IVL	a	✗
IVL_{RS}	b	✓

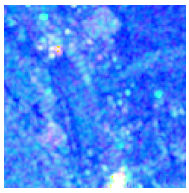


Reference map

Fig. 4: Qualitative examples for MCQ capabilities of considered models. The S2 input is visualized as the true color RGB representation of the S2 image and the S1 input is visualized as (RGB: VV, VH, VV/VH). We provide the reference maps as aid to the reader only. Reference maps are not provided as input to the model. "–" indicates that the answer could not be extracted.

Point out a continuous area of broad-leaved forest in the scene.

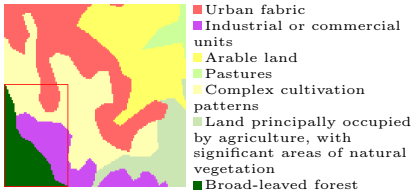
S2



S1

[0.0, 0.44, 0.35, 1.0]

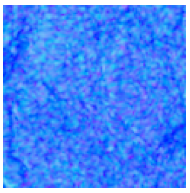
Model	Answer
GC	[0.0, 0.57, 0.24, 0.89]
LHRS	[0.45, 0.45, 0.89, 0.89]
SE	[0.65, 0.19, 0.83, 0.4]
ED_{rgb}	[0.0, 0.79, 0.21, 0.91]
ED_{s2}	–
EM_{rgb}	[0.08, 0.05, 0.52, 0.97]
EM_{s1s2}	[0.74, 0.25, 0.98, 0.63]
GPT	[0.02, 0.29, 0.34, 0.97]
Qwen	[0.0, 0.0, 0.17, 0.25]
GLM	[0.0, 0.0, 1.0, 1.0]
LLaVa	[0.0, 0.67, 0.21, 0.99]
IVL	–
IVL_{RS}	[0.0, 0.45, 0.35, 1.0]



Reference map

Output a bounding box surrounding the land cover class instance positioned at (0.92, 0.05) in the image.

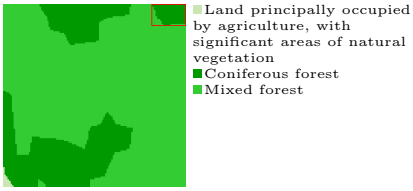
S2



S1

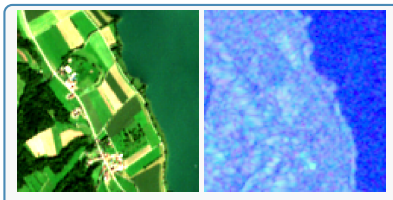
[0.81, 0.0, 1.0, 0.12]

Model	Answer
GC	[0.84, 0.0, 0.96, 0.2]
LHRS	[0.92, 0.05, 0.92, 0.05]
SE	–
ED_{rgb}	[0.06, 0.49, 0.1, 0.55, 0.0]
ED_{s2}	–
EM_{rgb}	[0.0, 0.23, 0.08, 0.92]
EM_{s1s2}	[0.86, 0.22, 0.86, 0.22]
GPT	[0.0, 0.0, 1.0, 1.0]
Qwen	[0.91, 0.04, 1.0, 0.08]
GLM	[0.85, 0.0, 1.0, 0.26]
LLaVa	[0.786, 0.054, 0.998, 0.308]
IVL	[0.93, 0.29, 0.94, 0.61]
IVL_{RS}	[0.8, 0.0, 1.0, 0.13]



Reference map

Fig. 5: Qualitative examples for referring expression detection capabilities of considered models. The S2 input is visualized as the true color RGB representation of the S2 image and the S1 input is visualized as (RGB: VV, VH, VV/VH). We provide the reference maps as aid to the reader only. Reference maps are not provided as input to the model. The instance that is to be detected is marked by a red rectangle. "–" indicates that the answer could not be extracted.



S2 S1

Instruction: Explain what this satellite image shows, mentioning the location, season, and landscape features.

Reference: This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant feature is arable land (770'000 sqm), which spans two areas, one large (∼761'000 sqm) and one marginal, covering a small fraction of the total area. The arable land borders a substantial body of inland waters (∼440'000 sqm) and a smaller patch of mixed forest (∼230'000 sqm). The landscape is characterized by a mix of agricultural areas, water bodies, and forest and seminatural areas, highlighting the varied ecosystems present in the region.

EarthMind (RGB) RGB

This satellite image from GoogleEarth shows a rural landscape with a mix of agricultural fields and forested areas. The fields are primarily located in the central and bottom left sections of the image, with a notable concentration of greenery. The forested areas are more prominent in the top left and bottom right sections, with a dense cluster of trees. A water body is visible in the top right corner, adding to the rural character of the scene. The image is taken in a season where the vegetation is in full, indicating a time of growth or harvest. The overall scene is a blend of agricultural land and natural forest, with a water feature adding to the rural setting.

EarthDial (RGB) RGB

The satellite image shows a large area of farmland and a dense forest near the coast. The farmland is located in the center of the image, while the forest is located at the bottom left. The coast is visible at the top right of the image. The image likely represents a coastal region with a mix of agricultural and natural landscapes.

GeoChat RGB

The satellite image shows a diverse landscape with a mix of urban and natural features. The center of the image displays a roundabout, which is likely a busy intersection in a city or town. Surrounding the roundabout are various buildings, indicating a dense urban area. In the background, there are trees and a roundabout, suggesting a well-balanced blend of urban and green spaces. The presence of a roundabout and trees together indicates that the location is likely a well-developed and well-maintained area.

SkyEyeGPT RGB

many farmlands are near a body of water

EarthMind (S1S2) S1,S2

This image shows a landscape with a mix of agricultural fields and forested areas. The fields are primarily located in the central and lower parts of the image, with a notable concentration in the bottom right. The forested areas are more prominent in the upper left and bottom left sections. The image is likely taken in a rural setting, possibly in a region with a combination of agricultural and natural land use. The presence of a water body on the right side suggests proximity to a river or lake, which could be a significant feature for the surrounding land use.

EarthDial (S2) S2

The satellite image shows a land mixed in the left of the image, water in the right of the image.

RS-InternVL RGB,S1,S2

The image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant feature is arable land, spanning approximately 787,000 square meters, which is distributed across two areas, one large (∼786,000 sqm) and one marginal. The arable land borders with inland waters (∼350,000 sqm), mixed forest (∼209,000 sqm), and urban fabric (∼72,000 sqm). The inland waters are adjacent to the mixed forest, which is spread over two areas, one smaller (∼199,000 sqm) and one marginal. The landscape presents a mix of agricultural areas, water bodies, and artificial surfaces, highlighting the varied ecosystems within this region.

LHRS-Bot RGB

Based on the provided image, the location appears to be a rural area with a mix of agricultural lands, forests, and water bodies. The image covers an area of approximately 144,000 square meters. Location:The image appears to be located in a temperate region, with a mix of flat and hilly terrain. The landscape features rolling hills, valleys, and a large lake in the center of the image. Season:The image appears to be captured during the summer season, with lush green vegetation and full crops. The trees are full of leaves, and the fields are filled with ripe crops, indicating a warm and sunny weather. Land Cover Types: 1. Agricultural lands: The image shows a mix of arable lands, pastures, and orchards. The arable lands are cultivated with various crops, including corn, soybeans, and wheat. The pastures are home to livestock, such as cows and sheep. The orchards are filled with fruit trees, such as apple and pear trees. 2. Forests: The image features a mix of deciduous and evergreen forests. The deciduous forests are dominated by trees such as oak, maple, and beech, which shed their leaves in the fall. The evergreen forests are home to coniferous trees such as pine, spruce, and fir. Water bodies: The image features a large lake in the center, surrounded by smaller ponds and streams. The lake is home to a variety of aquatic plants and animals, including fish, birds, and aquatic mammals. 4. Residential areas: The image shows a few small residential areas, likely consisting of single-family homes and farms. 5. Roads and infrastructure: The image features a network of roads, including a main road that runs through the center of the image. Other infrastructure, such as bridges, can be seen in the distance. 6. Recreational areas: The image shows a few small parks and recreational areas, likely used for activities such as hiking, fishing, and boating. Overall, the image depicts a rural landscape with a mix of agricultural lands, forests, water bodies, and residential areas. The season is likely summer, with lush green vegetation and full crops.

Fig. 6: A qualitative example of the captioning capabilities of considered RS models. The S2 input is visualized as the true color RGB representation of the S2 image and the S1 input is visualized as (RGB: VV, VH, VV/VH). The input modality for each model is noted in the top-right corner of the respective model answer.

InternVL3-1B RGB

The satellite image shows a rural landscape with distinct agricultural fields, indicating a countryside setting. Here's an explanation of what the image shows: 1. Location and Season: - The image area spans 144,000 square meters. - Given that it's a remote or aerial shot, it may have been taken in a time of year when the region is often in a different state of cultivation or growth, possibly indicating spring or summer. The vegetation seems lush, typical of a period when crops are actively harvested. 2. Landscape Features: - The landscape includes multiple fields of varying shades of green, with some sections showing more dense vegetation compared to others, possibly indicating different crops or stages of harvest. - There is a noticeable path or road cutting through the fields, running diagonally from the top left to the bottom right. This seems to be a main road, likely connecting different agricultural zones. - The fields are organized in strips and patterns, suggesting structured farming practices. The scattered structures among the fields could be farm buildings, barns, or storage facilities. - The surrounding area has dense greenery, indicating dense forests or possibly a wooded region. This helps in distinguishing the agricultural areas from the forested zones around them. - The water body to the right is likely a large lake, river, or inlet, typical of a riverine or estuarine setting. Its expansive size and darker color contrast with the green fields to the left, suggesting a significant body of water. Overall, the image captures a serene rural landscape with organized agricultural fields, a path for transportation, and natural surroundings suggesting a transitional period between different seasons or stages in agriculture.

GPT-5.2 RGB

A small rural area of patchwork farmland bordered by dense green woodland, likely in a temperate region during late spring or summer, showing rectangular crop fields in varying shades of green and tan, a few scattered farm buildings and houses connected by narrow roads, and sharp field boundaries and hedgerows across the roughly 1.44 km^2 scene.

LLaVA-OneVision RGB

The image shows a rural landscape with a variety of green fields, indicating a lush and fertile area. There is a body of water on the right side, which could be a river or a lake. The fields are divided into different sections, suggesting a mix of agricultural activities such as crop rotation or different types of farming. There are also some structures visible, which could be farmhouses or small buildings, indicating human habitation and activity in the area. The presence of trees and the overall greenery suggest that the season could be spring or summer. The image does not provide enough detail to determine the exact location or the specific types of crops being grown.

GLM-4.6-Flash RGB

The satellite image depicts a rural landscape with a mix of agricultural fields, residential structures, and a body of water (likely a lake or river). The location appears to be a countryside area, possibly in a temperate region given the green vegetation. The season is likely spring or summer, as the fields are lush and green, indicating active vegetation growth. Key landscape features include rectangular agricultural plots (suggesting organized farming), a cluster of buildings (residential or small-scale rural development), and a large water body to the right, which may serve as a natural boundary or resource for the area. The overall scene reflects a blend of human settlement and agricultural land use in a natural setting.

RS-InternVL RGB,S1,S2

The image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant feature is arable land, spanning approximately 787,000 square meters, which is distributed across two areas, one large (∼786,000 sqm) and one marginal. The arable land borders with inland waters (∼350,000 sqm), mixed forest (∼209,000 sqm), and urban fabric (∼72,000 sqm). The inland waters are adjacent to the mixed forest, which is spread over two areas, one smaller (∼199,000 sqm) and one marginal. The landscape presents a mix of agricultural areas, water bodies, and artificial surfaces, highlighting the varied ecosystems within this region.

Qwen3-VL RGB

Based on the satellite image and the provided area, here is an explanation of what the image shows: This satellite image displays a rural landscape located in the Czech Republic, specifically in the South Bohemian Region, near the town of Křešice (or possibly the nearby village of Křešice u České Lípy). The image captures a patchwork of agricultural fields adjacent to a large body of water, which is Lake Černé jezírko (Black Lake) or a similar local reservoir. Season: The image was likely taken during late spring or early summer. This is indicated by the vibrant green coloration of the fields, which suggests active growth of crops like cereals or vegetables. The absence of snow or bare soil rules out winter, and the lushness is more characteristic of spring/summer than autumn (when fields would be golden or harvested) or early winter. Landscape Features: - Agricultural Fields: The dominant feature is the grid-like pattern of cultivated land, divided into rectangular plots of varying shades of green and brown. These represent different crops or stages of growth, with some fields appearing darker green (possibly corn or lush grass) and others lighter or brownish (possibly wheat, barley, or fallow land).- Water Body: A large, dark green body of water occupies the right side of the image, indicating a lake or reservoir. Its smooth surface suggests calm conditions.- Roads and Settlements: A network of roads connects the fields and a small cluster of buildings visible near the center-left. These likely represent a rural village or hamlet.- Vegetation: Darker green patches, particularly along the edges of the fields and near the water, indicate forests or wooded areas, which are common in this part of the Czech countryside.- Topography: The terrain appears relatively flat, typical of the Bohemian lowlands, with slight variations in elevation suggested by the contours of the fields and the shoreline. In summary, the image shows a typical Czech rural landscape in late spring or early summer, featuring agricultural fields, a lake, and a small settlement, all within the South Bohemian Region. The 1,440,000 square meter area encompasses this entire scene, highlighting the interplay between agriculture, water, and human habitation.

Fig. 7: A qualitative example of the captioning task for considered CV models. The provided input is the same as in Fig. 6. The input modality for each model is noted in the top-right corner of the respective model answer.

H RS-InternVL Architecture Overview

The detailed architecture of RS-InternVL is provided in Fig. 8. We retain the original InternVL components (green) while adding additional paths for S1 and

S2 (orange). To leverage InternVL's pre-trained capabilities we provide the RGB bands of S2 to the InternViT. Only the projection layers and LoRA adapters for InternLM are fine-tuned.

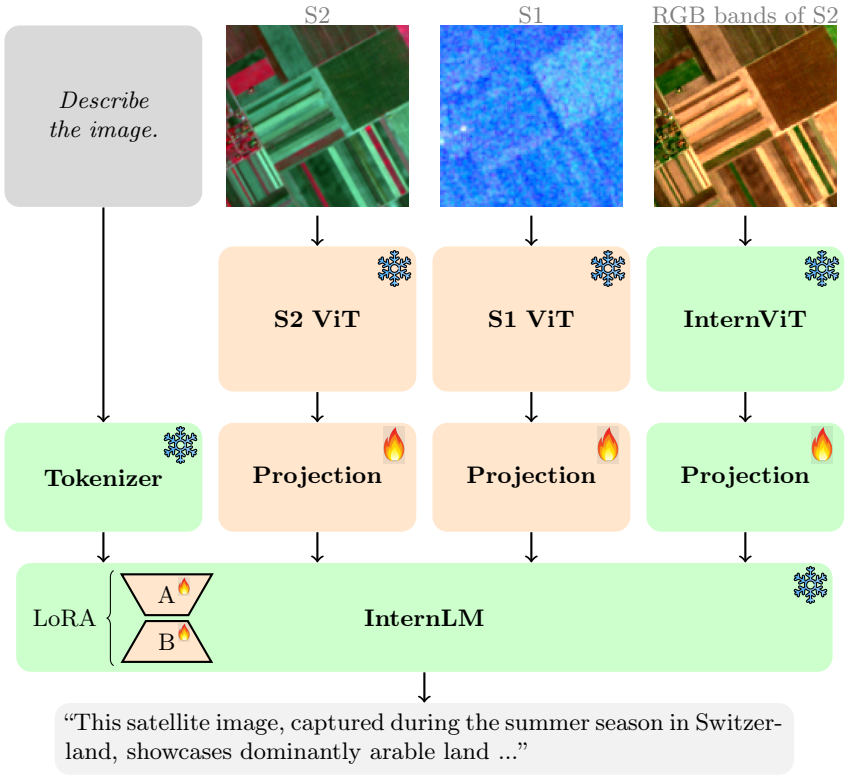


Fig. 8: Architectural overview of adapted InternVL to multi-sensor input. InternVL parts are marked in green while additions are orange. The S2 input is visualized as false color composite of an S2 image (RGB: Band 8 – NIR, Band 4 – Red, Band 3 – Green). The S1 input is visualized as (RGB: VV, VH, VV/VH).

I RS-InternVL Results for all Tasks

Tables 7 to 11 show results on the benchmark split for the fine-tuned RS-InternVL on all 15 tasks across the four categories binary VQA (Tab. 7), MCQ (Tab. 8), referring expression detection (Tabs. 9 and 10), and captioning (Tab. 11). We include the top result obtained by an RS model as well as a CV model for each task. One can clearly see that performance is increased drastically when adequate pre-training data is used. Only for the MCQ area task

Table 7: Results on the binary VQA tasks for VLMs in RS and CV: Presence, Area, Counting, Adjacency, and Overall, denoted as Pr, A, Cnt, Adj, and OA, in terms of accuracyfrom the fine-tuned RS-InternVL and best results from VLMs in RS and CV. All results are in percent (%).

Model	Pr	A	Cnt	Adj	OA
SOTA RS	69.34	56.20	51.99	60.62	57.90
SOTA CV	64.33	67.38	62.33	58.94	61.96
RS-InternVL	**69.68**	**82.39**	**78.02**	**68.81**	**73.29**

GPT performs slightly better, due to its much larger parameter count. Overall on MCQ GPT performs less favorably.

Table 8: Results on the MCQ tasks for VLMs in RS and CV: Presence, Area, Counting, Adjacency, Relative Position, Country, Season, Climate Zone, and Overall, denoted as Pr, A, Cnt, Adj, RP, Loc, S, Clt, and OA, respectively in terms of accuracy for the fine-tuned RS-InternVL and best results from VLMs in RS and CV. All results are in percent (%).

Model	Pr	A	Cnt	Adj	RP	Loc	S	Clt	OA
SOTA RS	46.40	27.41	36.43	44.70	31.60	54.17	25.71	37.04	35.26
SOTA CV	38.08	**44.42**	34.01	43.33	34.82	47.76	27.43	45.93	37.55
RS-InternVL	**64.88**	42.11	**55.20**	**56.90**	**57.98**	**88.78**	**89.14**	**88.44**	**65.84**

Table 9: Results on referring LULC detection for VLMs in RS and CV in terms of mIoU, and accuracy@k (where a prediction is counted as correct if the overlap with the reference is greater than k%). All results are in percent (%).

Model	mIoU	Acc@25	Acc@50	Acc@75	Acc@90
SOTA RS	22.65	38.41	17.88	5.67	1.39
SOTA CV	25.16	40.81	19.52	4.28	1.39
RS-InternVL	**33.16**	**47.73**	**34.38**	**18.77**	**6.55**

Table 10: Results on referring point detection for VLMs in RS and CV in terms of mIoU, and accuracy@k (where a prediction is counted as correct if the overlap with the reference is greater than k%). All results are in percent (%).

Model	mIoU	Acc@25	Acc@50	Acc@75	Acc@90
SOTA RS	12.65	17.51	5.08	1.27	0.13
SOTA CV	38.95	64.21	35.53	8.63	2.79
RS-InternVL	**69.95**	**95.43**	**82.23**	**48.22**	**12.69**

Table 11: Results on the captioning task for VLMs in RS and CV in terms of n-gram-based (BLEU-4, ROUGE, METEOR, CIDEr), sentence-embedding-based (BERTScore, SBert-Cosine) and LLM-based (CLAIR) metrics. All results are in percent (%).

Model	BLEU-4	ROUGE	METEOR	CIDEr	BERTScore	SBERT-Cosine	CLAIR
SOTA RS	1.66	16.54	14.93	0.96	84.34	51.97	59.70
SOTA CV	0.96	14.94	17.83	0.81	83.89	55.90	58.91
RS-InternVL	**34.04**	**50.63**	**50.62**	**78.35**	**91.87**	**83.29**	**70.90**